

性质

- memoryless: $y(t)$ 与 $x(t)$ 有关. eg. $y(t) = \cos 3t \cdot x(t)$ ✓ $\leftarrow$ 都与前项无关
- causal 因果: $y(t)$ 与 $x(t_0)$ 有关, $t_0 \leq t$. $y(t) = x(\frac{t}{2})$ $\times$ $y[n] = (\frac{1}{2})^n (x[n]+1)$ ✓ $x(s+nt)$ $\times$
- stable: $y(t)$ 有界 eg. 积分收敛 $\times$ $y[n] = (-\frac{1}{2})^n (x[n]+1)$ $\times$, $n \rightarrow \infty$ 复数看 $|y[n]|$ $x(s+nt)$ $\times$
- time-invariant 时不变: $X \rightarrow y$, $x(t+t_0) \rightarrow y(t+t_0)$ 前项 $f(t)$ $\times$, 积分收敛 $\checkmark$, $(t)(t)$ $\times$ $x(s+nt)$ $\times$
- linear $\int a x_1(t) + b x_2(t)$ 验证. ① $x(t)=0$ 时 $y(t)=0$. eg. $y[n] = (-\frac{1}{2})^n (x[n]+1)$ $\times$ 积分 $\checkmark$

基础

$$1+2+\dots+2^n = \frac{1-2^{n+1}}{1-2} \Rightarrow \frac{1}{1-2}$$

$$1+2+3+2^2+\dots = \frac{1}{(1-2)^2}$$

$$\cos \theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta}), \sin \theta = \frac{1}{2i}(e^{i\theta} - e^{-i\theta})$$

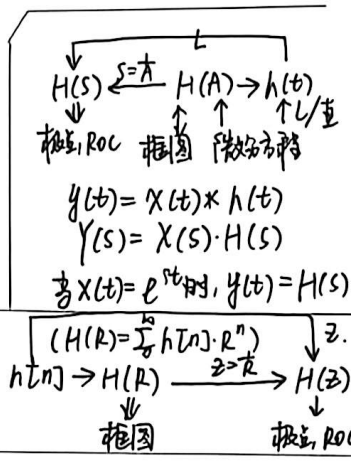
$$\delta[n] = u[n] - u[n-1]$$

$$\text{unit step } u[n], \text{ 若 } x[n] = u[n-b] - u[n-a] \begin{matrix} \uparrow & \uparrow \\ b & a \end{matrix}$$

$$\text{若 } x[n] = u[n-b] - u[n-a] \begin{matrix} \uparrow & \uparrow \\ b & a \end{matrix}$$

Discrete-time 离散 Continuous-time 连续

- Laplace $H(s) = \int h(t) e^{-st} dt$
- Block diagram
- System functional: $H(A)/H(E)$
- Impulse response: $h(t)$ / 卷积
- Differential equation 微分方程
- System function $H(s)/H(z)$
- step response $u(t)$ 输入输出 $y(t)$
- z变换 $H(z) = \sum h[n] \cdot z^{-n}$
- unit-sample response: $h[n]$
- difference equation 差分方程



how $H(A) \rightarrow h(t)$:

$$H(A) = \frac{2A}{2+3A+A^2} = \frac{2A}{(1+A)(1+2A)} = \frac{2A}{1+A} - \frac{2A}{1+2A} = 2A(1-A+\dots) - 2A(1-2A+\dots)$$

$$X = \delta(t), AX = u(t) \Rightarrow Y = 2u(t) [(1-\frac{1}{2}t+\dots) - (1-t+\dots)] = 2(e^{-\frac{1}{2}t} - e^{-t}) u(t).$$

因果系统: $h(t)$ 右边信号, ROC 在最右侧极点之右
 稳定系统: ROC 包含虚轴 因果稳定系统: 极点都在虚轴左边

因果: ROC 在外国. 稳定: ROC 包含单位圆.
 因果稳定: 所有极点都在圆内

计算

L: $e^{-at} \rightarrow \frac{1}{s+a}$ $\delta(t) \rightarrow 1$
 $\cos t \rightarrow \frac{s}{s^2+1}$ $u(t) \rightarrow \frac{1}{s}$
 $\sin t \rightarrow \frac{1}{s^2+1}$ $t u(t) \rightarrow \frac{1}{s^2}$
 $x(at) = \lim_{s \rightarrow \infty} s X(s)$
 $x(0) = \lim_{s \rightarrow 0} s X(s)$

时域: $y(t) \rightarrow s Y(s)$ $e^{-t} u(t) \rightarrow \frac{1}{s+1}$
 $y(t-t_0) \rightarrow e^{-st_0} Y(s)$ $e^{-t} u(-t) \rightarrow -\frac{1}{s-1}$
 $t y(t) \rightarrow -\frac{d}{ds} Y(s)$ $\int_0^t y(\tau) d\tau \rightarrow \frac{1}{s} Y(s)$
 $y(-t) \rightarrow Y(-s)$
 $e^{st_0} y(t) \rightarrow Y(s-s_0)$
 $y(at) \rightarrow \frac{1}{a} Y(\frac{s}{a})$

z: $\delta[n] \rightarrow 1$ $n x[n] \rightarrow -z \frac{d}{dz} X(z)$
 $u[n] \rightarrow \frac{z}{z-1}$ $x[n-n_0] \rightarrow z^{-n_0} X(z)$
 $n u[n] \rightarrow \frac{z}{(z-1)^2}$ $2^n u[n] \rightarrow \frac{z}{z-2}$
 $-n u[-n-1] \rightarrow \frac{1}{(z-1)^2}$ $-2^n u[-n-1] \rightarrow \frac{1}{z-2}$
 $\delta[n-1] \rightarrow z^{-1}$ $n 2^n u[n] \rightarrow \frac{dz}{(z-2)^2}$
 $a^n x[n] \rightarrow X(\frac{z}{a})$
 $x[-n] \rightarrow X(z^{-1})$

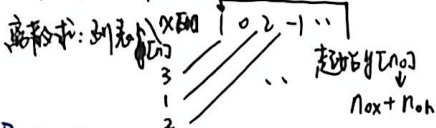
一些思路技巧

$x(t) = \sum_{n=0}^{\infty} (\frac{1}{2})^n (u(t-n) - u(t-n-\frac{1}{2}))$
 看到 $\square$, 高项相减.
 $\Rightarrow X(s) = \int_0^{\infty} \sum_{n=0}^{\infty} (\frac{1}{2})^n (u(t-n) - u(t-n-\frac{1}{2})) e^{-st} dt$
 $= \sum_{n=0}^{\infty} (\frac{1}{2})^n \int_n^{n+\frac{1}{2}} e^{-st} dt$
 $u(t-n)$ 积分范围: $\int_n^{n+\frac{1}{2}} e^{-st} dt$
 $x \rightarrow \text{block} \rightarrow y \leftarrow H(s) = \frac{1}{s+2}$
 $x_e[n] = \frac{1}{2}(x[n] + x[-n]), x_o[n] = \frac{1}{2}(x[n] - x[-n])$
 $\text{令 } m(t) = t u(t)$ 斜率改变一次.
 看到 $e^{-st_0} \cdot \frac{1}{s^2}$
 $\frac{s-1}{(s+1)^2} \rightarrow \frac{1}{2} \frac{1}{(s+1)^2} + \frac{1}{2} \frac{1}{(s+1)} \Rightarrow t \cos t u(t)$
 或看成 $(\frac{s-1}{s+1})'$
 卷积遇到 $\frac{1}{a} \frac{1}{b}$ 不对称 $u(-t) - u(t)$, 直接 $\int_a^b$

卷积 Convolution

$$DT: y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k]$$

$$CT: y(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$$



$$时域: x * h = h * x$$

$$x * (h_1 + h_2) = x * h_1 + x * h_2$$

$$x * h_1 * h_2 = x * (h_1 * h_2)$$

$$x * \delta(t) = x, x * \delta(t-t_0) = x(t-t_0)$$

$$\text{Step response } S(t) = \int_{-\infty}^t h(\tau) d\tau$$

$$t \neq 0 \Leftrightarrow h(t) = 0$$

$$\text{因果: } t < 0 \text{ 时 } h(t) = 0$$

$$\text{稳定: } \int_{-\infty}^{\infty} |h(t)| dt < \infty$$

Symmetry 对称

Bode diagram

① magnitude: $\uparrow \text{dB} = 20 \lg |H(j\omega)|$

$$w=0 \text{ 时, 增益 } = 20 \lg K$$

$$H(j\omega) = K \cdot \frac{(1+j\frac{\omega}{z_1}) \dots}{(1+j\frac{\omega}{p_1})(1+j\frac{\omega}{p_2}) \dots}$$

零极点: 斜率增加 20 dB/dec

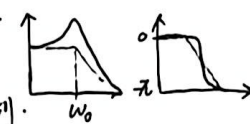
极点: 斜率减少 20 dB/dec

若分母有 $j\omega/s$: 斜率增加 20 dB/dec. eg $H(s) = s$:

② angle: $\begin{cases} \text{零极点 } (1+j\frac{\omega}{z_1}): \text{在 } [\frac{\omega}{z_1}, 10\frac{\omega}{z_1}] \text{ 内增加 } 90^\circ \\ \text{极点 } (1+j\frac{\omega}{p_1}): \text{在 } [\frac{\omega}{p_1}, 10\frac{\omega}{p_1}] \text{ 内减少 } 90^\circ \end{cases}$

③ 高通系统: $H(s) = \frac{s}{s^2 + \dots} \rightarrow +40 \text{ dB/dec, } +180^\circ$ (若是右半平面零极点, -180°)

$$H(s) = \frac{1}{1 + \frac{s}{\omega_0} + (\frac{s}{\omega_0})^2}$$



矢量图

$$H(s) = K \frac{(s-z_1) \dots}{(s-p_1) \dots} \Rightarrow |H(s_0)| = |K| \cdot \frac{\prod |s_0 - z_i|}{\prod |s_0 - p_i|}, \angle H(s_0) = \angle K + \sum \angle (s_0 - z_i) - \sum \angle (s_0 - p_i)$$



	离散傅里叶级数 Fourier Series (周期)	傅里叶变换 Fourier Transform (非周期)	时域 → 频域
离散 DT	$X[n] = \sum_{k=-\infty}^{\infty} a_k e^{j\frac{2\pi}{T}kn}, T=N$ $a_k = \frac{1}{N} \sum_{n=-\infty}^{\infty} X[n] e^{-j\frac{2\pi}{T}kn}, T=N$	$X(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} x(t) e^{j\omega t} dt$ $x(t) = \int_{-\infty}^{\infty} X(\omega) e^{-j\omega t} d\omega$	连续 → 非周期 离散 → 周期 ($T=2\pi$) 周期 → 离散 非周期 → 连续
连续 CT	$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{j\frac{2\pi}{T}kt}$ $a_k = \frac{1}{T} \int_T x(t) e^{-j\frac{2\pi}{T}kt} dt, T$	$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$ $x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$	对 DT, $X[n]$ 以 T 为周期 $\Rightarrow a_k$ 以 T 为周期

性质	CTFS: $x(t) \rightarrow a_k$	CTFT: $x(t) \rightarrow X(\omega)$	DTFT: $x[n] \rightarrow X(e^{j\omega})$
时移	$x(t-t_0) \rightarrow a_k e^{-jK\omega_0 t_0}$	$x(t-t_0) \rightarrow X(\omega) e^{-j\omega t_0}$	$x[n-n_0] \rightarrow X(e^{j\omega}) e^{-j\omega n_0}$
频移	$e^{jK\omega_0 t} x(t) \rightarrow a_{k-K}$	$e^{j\omega_0 t} x(t) \rightarrow X(\omega - \omega_0)$ 右移	$e^{j\omega_0 n} x[n] \rightarrow X(e^{j(\omega - \omega_0)})$, $\omega_0 = 2\pi/T$
共轭	$x^*(t) \rightarrow a_{-k}^*$	$x^*(t) \rightarrow X^*(-\omega)$	
$x(t)$ 实函数时	$a_k = a_{-k}^*$	$X(\omega) = X^*(-\omega)$	
时域微分	$x'(t) \rightarrow jK\omega_0 a_k$	$x'(t) \rightarrow j\omega X(\omega)$	
时域积分	$\int_{-\infty}^t x(\tau) d\tau \rightarrow \frac{a_k}{jK\omega_0}$	$\int_{-\infty}^t x(\tau) d\tau \rightarrow \frac{X(\omega)}{j\omega} + \pi X(0) \delta(\omega)$	
	$\frac{1}{T} \int_T x(t) ^2 dt = \sum_{k=-\infty}^{\infty} a_k ^2$	$\int_{-\infty}^{\infty} x(t) ^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) ^2 d\omega$	$\sum_{n=-\infty}^{\infty} x[n] ^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) ^2 d\omega$
	$x(2t) \rightarrow a_k$ 不变, $2>0$	$x(2t) \rightarrow \frac{1}{ 2 } X(\frac{\omega}{2})$	$x[\frac{n}{2}] \rightarrow X(j\omega K)$
即: $X_0(t) \rightarrow \text{Im}$, $X_e(t) \rightarrow \text{Re}$		$t x(t) \rightarrow j \frac{d}{d\omega} (X(\omega))$	$n x[n] \rightarrow j \frac{d}{d\omega} (X(e^{j\omega}))$
$x(t)$ 奇函数 $\text{Re}(a_k)=0$		$f(t) \rightarrow F(\omega)$, 则 $F(jt) \rightarrow 2\pi f(-\omega)$	$\sum x_e(t) = \frac{1}{2} [x(t) + x(-t)]$
$x(t)$ 偶 $\text{Im}(a_k)=0$		$\text{Re}(X(\omega))=0$	$X_0(t) = \frac{1}{2} [x(t) - x(-t)], t>0$
		$\text{Im}(X(\omega))=0$	
		$\int x(t) y(t) dt \rightarrow \frac{1}{2\pi} X(\omega) * Y(\omega)$	$\delta[n] \rightarrow 1$, $u[n] \rightarrow \frac{1}{1-e^{-j\omega}}$
		$x(t) * y(t) \rightarrow X(\omega) Y(\omega)$	$1 \rightarrow 2\pi \sum_{k=-\infty}^{\infty} \delta(\omega - 2k\pi)$

傅里叶变换对: $x(t) \rightarrow X(\omega)$

- $\delta(t) \rightarrow 1$
- $1 \rightarrow 2\pi \delta(\omega)$
- $e^{-at} u(t) \rightarrow \frac{1}{a+j\omega}$
- $\text{rect}(t) \rightarrow \text{sinc}(\omega)$
- $\frac{\sin \omega_0 t}{\pi t} \rightarrow \text{rect}(\omega)$
- $\cos(\omega_0 t) \rightarrow \pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$
- $\sin(\omega_0 t) \rightarrow \pi [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$
- $u(t) \rightarrow \frac{1}{j\omega} + \pi \delta(\omega)$

傅里叶级数对: $x(t) \rightarrow a_k$

- $\delta(t) \rightarrow 1$
- $1 \rightarrow 2\pi \delta(\omega)$
- $e^{-at} u(t) \rightarrow \frac{1}{a+j\omega}$
- $\text{rect}(t) \rightarrow \text{sinc}(\omega)$
- $\frac{\sin \omega_0 t}{\pi t} \rightarrow \text{rect}(\omega)$
- $\cos(\omega_0 t) \rightarrow \pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$
- $\sin(\omega_0 t) \rightarrow \pi [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$
- $u(t) \rightarrow \frac{1}{j\omega} + \pi \delta(\omega)$

对连续 关系变换

非周期: a_k $\rightarrow$ 周期: a'_k $\rightarrow$ 同: a'_k $\rightarrow$ 延拓

$X(\omega) \rightarrow a'_k = \frac{1}{T} X(jK\omega_0)$ $\rightarrow$ 相当于: ω_0 采样 + $\frac{2\pi}{T}$ 幅度

$X'(j\omega) = 2\pi \sum_{k=-\infty}^{\infty} a'_k \delta(\omega - k\omega_0)$ $\rightarrow$ 周期: $b_k = a_k \cdot H(jK\omega_0)$, $\omega_0 = \frac{2\pi}{T}$

非周期: $Y(\omega) = X(\omega) H(\omega)$ (离散也一样)

问是否 LTI: 看 a_k 和 b_k , 看 $a_k=0$ 时若 $b_k \neq 0 \Rightarrow X$

基础知识

$$\cos \theta = \frac{1}{2}(e^{j\theta} + e^{-j\theta})$$

$$\sin \theta = \frac{1}{2j}(e^{j\theta} - e^{-j\theta})$$

看到 $\Delta \Rightarrow$ 两个矩形的卷积:

看到 $\Delta \Rightarrow$ 两个矩形的卷积: $X(j\omega) * \delta(\omega - \omega_0) = X(j\omega_0)$, $\int_{-\infty}^{\infty} f(x) \delta(x - x_0) dx = f(x_0)$ 相乘, 采样.

看到 $\Delta \Rightarrow$ $A_T = \frac{\sin(K\omega_0 T)}{K\omega_0 T}$

看到 $\Delta \Rightarrow \sum_{n=-\infty}^{\infty} e^{jn\omega_0 t} = T \sum_{n=-\infty}^{\infty} \delta(t - nT)$, $T = \frac{2\pi}{\omega_0}$

看到 $\Delta \Rightarrow$ $X[n] = 4\delta[n-1] + 4\delta[n+1]$ 拖 $T=8$ 延拓: $X[n] = \sum_{k=-\infty}^{\infty} 4\delta[n-1+8k] + 4\delta[n+1+8k]$

看到 $\Delta \Rightarrow$ $X = |X| \cdot e^{j\angle X}$ 分离出表达式 (注意反相而幅度是 180°).

看到 $\Delta \Rightarrow$ $\square - \Delta$

关于 Δ : $2\pi \cdot \frac{A}{a} \cdot \left(\frac{\sin(\frac{a}{2}\omega)}{\omega}\right)^2 \xrightarrow{F} \Delta$
 $\Delta \rightarrow A \cdot \left(\frac{\sin(\frac{a}{2}\omega)}{\frac{a}{2}\omega}\right)^2$

例: $|X(j\omega)|$

例: $X'(t) \rightarrow j\omega X(j\omega)$

$X * X(t) \rightarrow X^2$

$X(t - \frac{\pi}{2}) \rightarrow X(j\omega) \cdot e^{-j\omega \frac{\pi}{2}}$ $|X|$ 不变

$X(\pi t) \rightarrow \frac{1}{\pi} X(j\frac{\omega}{\pi})$

$X^2(t) \rightarrow \frac{1}{2\pi} X * X$ $\angle$ 不变

例: $X(e^{j\omega}) = \sin \frac{\omega}{2} \cdot e^{j(\frac{\pi}{2} - \frac{\omega}{2})} = -j \sin \frac{\omega}{2} \cdot e^{-j\frac{\omega}{2}}$

例: $X(j\omega)$
 若 $w(t) = \sum_{n=-\infty}^{\infty} X(t) \delta(t - 3n)$, 则 $w(t) = X(t) \cdot \sum_{n=-\infty}^{\infty} \delta(t - 3n)$

$P(j\omega) = \sum_{k=-\infty}^{\infty} \delta(\omega - k \cdot \frac{2\pi}{3})$ 平移.
 $W(j\omega) = \frac{1}{2\pi} X(j\omega) * P(j\omega) = \frac{1}{3} X(j\omega) \sum_{k=-\infty}^{\infty} \delta(\omega - k \cdot \frac{2\pi}{3})$

例: $b_k = \begin{cases} 1, & k \text{ odd} \\ 0, & k \text{ even} \end{cases}$ 求 $x(t)$, $(T=4)$

① $x(t) = \sum_{k=-\infty}^{\infty} e^{j\frac{\pi}{2}kt}$, 令 $k=2n+1$

$\Rightarrow x(t) = \sum_{n=-\infty}^{\infty} e^{j\frac{\pi}{2}(2n+1)t} = e^{j\frac{\pi}{2}t} \sum_{n=-\infty}^{\infty} e^{j\pi nt} = j^t \cdot 2 \sum_{n=-\infty}^{\infty} \delta(t - 2n) \stackrel{(-1)^n}{=} \sum_{n=-\infty}^{\infty} \delta(t - 2n)$

$\therefore x(t)$ 周期, 求 $x(t)$ 在 $t \in [0, 4)$ 内:

$n=0, 1: x(t) = 2\delta(t) - 2\delta(t-2)$, $t \in [0, 4)$.

② $x(t) = \sum_{k=-\infty}^{\infty} e^{j\frac{\pi}{2}kt} - \sum_{k=-\infty}^{\infty} e^{j\frac{\pi}{2}kt} = \sum_{k=-\infty}^{\infty} e^{j\frac{\pi}{2}kt} - \sum_{k=-\infty}^{\infty} e^{j\frac{\pi}{2}kt}$

$\Rightarrow [4\delta(t-4n) - 2\delta(t-2n)]$

$n=0, 1: (4\delta(t) - 2\delta(t)) - 2\delta(t-2) = 2\delta(t) - 2\delta(t-2)$

$$\sin 2 \cos \beta = \frac{\sin(2+\beta) + \sin(2-\beta)}{2}$$

$$\cos 2 \cos \beta = \frac{\cos(2+\beta) + \cos(2-\beta)}{2}$$

$$\sin 2 \sin \beta = \frac{-\cos(2+\beta) + \cos(2-\beta)}{2}$$

$$1 + 2 + \dots + 2^n = \frac{1 - 2^{n+1}}{1 - 2}$$

$$\delta[n] = u[n] - u[n-1]$$

$$\chi(t) = u(t-b) - u(t-a)$$

$$\chi[n] = u[n-b] - u[n-a]$$

卷积定义:

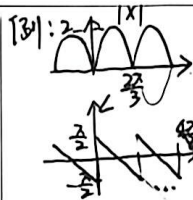
$$DT: y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k]$$

$$CT: y(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$$

inverse 反.

symmetric 对称

aliasing 混叠.



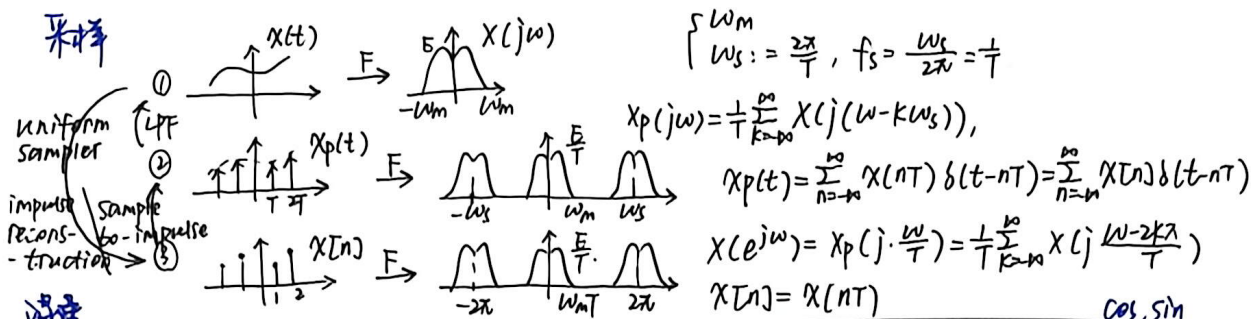
求 $H(j\omega)$:
 $|X|$ 翻转 $\Rightarrow \angle$ 上下移 π .
 $\angle$ 上下移 2π 等效

$H = 2 \sin(\frac{3}{2}\omega) e^{j(\frac{\pi}{2} - \frac{3}{2}\omega)}$
 $= 2j \sin(\frac{3}{2}\omega) \cdot e^{-j\frac{3}{2}\omega}$
 $= 1 - e^{-j3\omega}$



扫描全能王 创建

采样



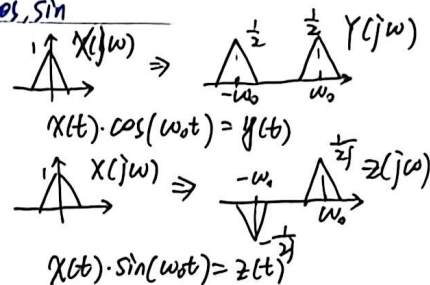
混叠

采样定理: 对 $x(t)$, 用 $\omega_s > 2\omega_m$ 采样, 得 $x[n] = x(nT)$ 可唯一恢复 $x(t)$

对任意DT信号, $x[n]$, $T=2\pi$. 先映射到 $[0, \pi]$, 越靠近0频率越低, 越靠近 π 频率越高.

对 $x[n](f)$, 经过采样后, 即如有混叠, 看成 $f_{out} = |f - m f_s|$

Cos, Sin



例: 已知 $x(t)$, $y(t)$. 是否存在 $h(t)$ 使其是 LTI.

① 算 $X(j\omega)Y(j\omega)$ 或 $a_k b_k$, 若 $X/a=0$ 且 $Y/b \neq 0$ 就不行.

② 直接看零极点. eg. (1) $x(t) = \sin(2\pi \cdot 100t) \Rightarrow \omega = 200\pi$.

$T=0.11, \omega_0 = \frac{2\pi}{T} = \frac{2\pi}{0.11} \Rightarrow \omega$ 是 ω_0 的整数倍. 可以

若 $x(t)$ 有 T , 则 $X(j\omega)$: 只在 $k\omega_0$ 上有分量.

$x(t)$ 周期 $\begin{cases} T$ 内: 决定 $X(j\omega)$ 包络
 T : 决定冲激

(2) $x_4(t) T=0.15$, T 内部是 $\frac{\sin(\pi \cdot \frac{1}{0.006})}{\pi \cdot \frac{1}{0.006}}$, $y(t)$ 同上.

$\omega_0 = 20\pi$, ω 是 ω_0 的整数倍.

但: 对方滤波器, $\frac{1}{0.006} = 167\pi < 200\pi$ 在 200π 处 $\omega = 0$. 不可以

求方法: ① $x(t)/x[n]$ 周期: 求 $H(j\omega)$, $a_k \Rightarrow b_k$

② 非周期: 求 $H(j\omega)$, $X(j\omega) \Rightarrow Y(j\omega)$ ($Y(j\omega)$ 算出来不好看时, 求 $x(t)h(t)$ 直接卷积) (牛特别有办法).

例: $x(t) T=1$, 用 $\Delta T = \frac{1}{6}$ 采样 $\Rightarrow x[n]$. 已知 $x[n]$: 偶数 ≥ 0 ; $x[0] + \dots + x[5] = 3$; $x[0] - x[1] + x[2] - x[3] + x[4] - x[5] = 1$; $a_0 - a_5$ 中只有2个非0. 给出2个不同 $x(t)$.

$a_k = \frac{1}{6} \sum x[n] e^{jk\frac{2\pi}{6}}$, $\Rightarrow a[0] = \frac{3}{6} = \frac{1}{2}$, $a[3] = \frac{1}{6}$. $x[n] = \frac{1}{2} + \frac{1}{6} e^{j\lambda n} = \frac{1}{2} + \frac{1}{6} (-1)^n = \frac{1}{2} + \frac{1}{6} \cos(\lambda n) = \frac{1}{2} + \frac{1}{6} \cos(3\lambda n)$

$t = \frac{n}{6} \Rightarrow n = 6t$, $x[n] = \frac{1}{2} + \frac{1}{6} \cos(\pi \cdot 6t) = \frac{1}{2} + \frac{1}{6} \cos(6\pi t) = \frac{1}{2} + \frac{1}{6} \cos(18\pi t)$.

$u[n] \xrightarrow{h[n]} s[n] = \sum_{k=-\infty}^{\infty} h[k] u[n-k]$, $s(e^{j\omega}) = H(e^{j\omega}) \cdot \frac{1}{1-e^{j\omega}}$

例: $y[n] - \frac{3}{4}y[n-1] + \frac{1}{8}y[n-2] = x[n]$.

$\Rightarrow Y(e^{j\omega})[1 - \frac{3}{4}e^{-j\omega} + \frac{1}{8}e^{-2j\omega}] = X(e^{j\omega}) \Rightarrow H(e^{j\omega}) = 4(\frac{1}{2-e^{j\omega}} - \frac{1}{4-e^{j\omega}})$

$\Rightarrow h[n] = 4(\frac{1}{2}(\frac{1}{2})^n u[n] - \frac{1}{4}(\frac{1}{4})^n u[n]) = [2(\frac{1}{2})^n - (\frac{1}{4})^n] u[n]$.

